Student Entrepreneurs for Economic Development (SEED)

University of Virginia

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Final Deliverable: Case Study of Nigerian Economic Data

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Meet the Team

Jared Orr is a Project Director at SEED majoring in Finance and Accounting with a minor in Entrepreneurship in UVA's McIntire School of Commerce. Compelled toward quantitative analysis and strategy, Jared will be interning as an Investment Banking Summer Analyst with Fifth Third Securities in 2021.

Caitlin Murphy is the Project Leader on the FinMango consulting project and is from New York City, NY. She is currently in UVA's Batten School studying Public Policy and Leadership along with Mathematics. As a creative and innovative individual, she looks forward to applying her skills and interests to enter the nonprofit sphere after graduating.

Alexander Mitchell is a Project Member on the FinMango consulting project and is from Arlington, VA. He is a second-year intending on studying economics and commerce. He is interested in pursuing a career in finance or consulting and has a passion for problem solving and helping others.

Bailey Mohri is a Project Member from Pittsburgh, PA. He is currently a second year intending on double majoring in commerce and psychology. Having exhibited his own artwork in the Andy Warhol Museum, he utilizes creativity in solution generation and idea communication– skills he hopes to use in a future consulting career.

Harsha Haribhaskar is a Project Member on the FinMango consulting project and is from Chennai, India. He is a second year intending on majoring in Commerce and Statistics. He has a passion for data science and development economics, and hopes to pursue a career in consulting relating to these fields in the future.

Leigh Mante is a Project Member on the FinMango consulting project. She is a third year from Virginia Beach, VA majoring in Global Public Health and Statistics and minoring in Global Sustainability. She is interested in pursuing a career at the intersection of international development and data science.

Nathan Manthei is a Project Member on SEED's FinMango consulting project. He is intending on studying Finance and Accounting in the McIntire School of Commerce. Outside of school, Nathan loves to play soccer and has interests in financial markets and the business of healthcare.

Executive Summary

In partnership with FinMango, our Student Entrepreneurs for Economic Development (SEED) at the University of Virginia team gathered, framed, and analyzed key economic data indicators for Nigeria. The following pages document our approach to collecting relevant data, structuring this data within our chosen indicators, and reporting the significant Nigerian data collection obstacles.

Given the difficulties associated with the international reporting disparities and economic effects of the pandemic, our team encountered several challenges in enacting our data collection methodology process. These namely include:

1. A decentralized reporting system of COVID-19 data,
2. An often obsolete selection of data for certain Nigerian states, and
3. An insufficient platform of reporting institutions, leading to scattered results.

Our procedure and resulting analysis reaffirm FinMango's mission to collect and present data of the pandemic's effects on the economics of various African countries. The analysis and resulting impacts of this effort remain bold and relevant given the uneven effect of the pandemic on various developing countries across the world. Through advocating for centralized reporting and achieving efficient automation of data collection, FinMango can extrapolate this methodology for other global regions, including Latin America and Asia Pacific.

On behalf of our entire team and the SEED organization as whole, we appreciate the opportunity to collaborate with FinMango this fall semester. We look forward to hearing your feedback and hopefully participating in your efforts this upcoming spring. If you have any questions, please contact Jared Orr at jeo5qp@virginia.edu and Caitlin Murphy at ccm8fy@virginia.edu.

Introduction: FinMango and Economic Data Collection Project

FinMango is a non-profit organization that uses data, technology, and advocacy to provide practical financial education to empower developing communities around the world. Having impacted more than 360,000 people, FinMango continues to work toward ending the global financial literacy crisis. The organization partners with a wide range of financial institutions, businesses, and governments to build financial inclusion projects around the world that are not only sustainable but also originated directly from the communities themselves. With a diverse group of forward-thinking volunteers, FinMango addresses real-world problems with practical solutions and provides hope for a better life.

FinMango has recently branched out to focus on the power of data and technology in developing a better understanding of contemporary global health issues, notably during the COVID-19 pandemic. Collaborating with Google's Open Data Project, FinMango contributes toward creating a comprehensive COVID-19 dataset. The goal of this dataset is to better understand the impact of the virus on different communities, provide unbiased data about the pandemic, and form partnerships between data providers and users to obtain insights and make critical decisions.

Prior to our time with FinMango, the dearth of available COVID-19 data challenged FinMango's data collection efforts, since much of the data was decentralized or published in formats difficult to aggregate, automate, or analyze, such as graphs published as image files or displayed using Tableau.

Currently, FinMango is still solving this problem by manually inputting data into an accessible repository for data from several African nations. We enjoyed freedom concerning what country or nation that the team would ultimately explore in depth, and because of the great variety of topics such as socioeconomic standards, healthcare access and quality, we focused our efforts on exploring Nigeria. In particular, this semester with FinMango was devoted to assisting the nonprofit primarily through mining and analyzing data to further examine the impact of COVID-19 across the individual states within the nation.

Data Collection Process

One of the most significant hurdles we faced in gathering data was the decentralized nature of COVID-19 data, a challenge similarly experienced by FinMango. While some sources contained data on certain states within Nigeria, they often did not have data for other states. For some regions, like the Federal Capital Territory and Enugu, no data existed on the websites we used to source data for most of the other states. Especially regarding COVID-19 data collection, a large part of data was represented using graphs on websites, which were only accessible as images. Consequently, the data points used in the construction of the graph were not available to us. FinMango has created a data repository for Kenya and South Africa, where data can be manually entered; however, this method is not feasible for many other countries in Africa, where data may not be as readily available.

A statement released by the United Nations Office of the Special Advisor on Africa (OSAA) said that the UN and the African Union will work together to overcome the challenge of obtaining the resources necessary to combat the direct and indirect non-health consequences of COVID-19. If FinMango wants to contribute to the UN's efforts to interpret the socioeconomic effects of COVID-19, it is critical to establish centralization for African data, especially at the local and regional level.

We identified several fundamental indicators essential to understanding the socioeconomic effects of COVID-19 in Africa; these included the GDP, the proportion of the population below the poverty line, and the proportion of the workforce that were male and female. We manually input the data into our dataset, using several sources to obtain the data for our selected indicators. The Nigerian Bureau of Statistics and data platforms like Knoema were especially useful in collecting data for these indicators. However, these sources did not contain data for all the indicators we had identified.

Another issue we had with data collection was that some of our sources had data that was outdated by ten or more years. Unfortunately, since there were few reliable alternatives to these sources, we had to use this old and incomplete data. For example, we were able to find only one reliable source that reported a GDP for the Federal Capital Territory in Nigeria; however, the amount was more than ten times the GDP of Lagos, the most populous city not only in Nigeria, but in the entire African continent. Lagos is considered to be one of the fastest growing cities in the world and has one of the highest GDPs in Africa.

These issues highlight the need for a centralized African data population, as such a platform will have the resources to regularly collect data every few years, if not annually. A centralized platform will also be more reliable, as it will:

- Be operated by a government agency, an organization contracted by the government, or a well-regarded NGO, and
- Have data that can be compared across the board, instead of having contradictory data from two states. Having such credentials means that data from these sources will be considered more reliable.

We also automated some parts of the data inputting process. We wanted to convert the GDP of each state in Nigeria from Nigerian Naira to US Dollars. Instead of calculating these conversions manually or by using a currency conversion calculator, we used the “Googlefinance” function in Google Sheets to convert the Nigerian Naira value to US Dollars quickly. Furthermore, we would be far more likely to make mistakes when inputting manually converted values, especially when dealing with decimals; automating this process leads to minimal error. Finally, the “Googlefinance” function also helps keep the data up-to-date, as it updates the values based on the current currency exchange rate. This function saved us a considerable amount of time, and it emphasizes the power of automation in making data inputting faster, updated, and more accurate.

Case Study of Nigeria

In selecting our African country of focus, we pursued Nigeria, as the country has a relatively strong economy, is more developed than other African nations, and the official language is English. Also, Nigeria responded quickly to the COVID-19 pandemic, instituting a strict lockdown in April.

We looked at a variety of different measures of the overall economic well being in Nigeria for each of the country’s 36 states. We analyzed GDP measures, income data, poverty statistics, and employment numbers for each state. Throughout the process, we succeeded in finding data for all of these categories except for income data. In addition, some data was missing for certain states. For example, there was no poverty or employment data for Borno, due to the statewide insecurity resulting from the presence of Boko Haram.

One challenge we encountered was finding accurate and up-to-date data for each of the Nigerian states. The only GDP data that we could find was from 2007 from a database named Knoema. This outdated data is likely not very useful for measuring a state’s current level of economic activity because much has likely changed in the last 13 years. However, the remainder of our data is much more recent than the GDP data. The poverty data and a portion of the employment data are both from 2019. The poverty data is sourced from a publication of the Nigerian government on poverty and inequality in Nigeria. The portion of the employment data from 2019, which is the data including measures of employment by gender, is sourced from a Nigeria living standards survey. The remaining portion of the employment data, which includes measures of the population workforce and the number of people employed, is from an unemployment report published by the Nigerian government covering the second quarter of 2020.

Encountered Challenges

Throughout the development of our data aggregation strategy, we observed a number of challenges. As mentioned previously, a significant difficulty was the ability to locate data regarding GDP by state. While the majority of the Nigerian dataset we were able to compile was current (2018-2020), GDP by state in local currency was an indicator lacking current information, with the most recent information being updated in 2007. Even then, the GDP was reported in international dollars using purchasing power parity rates unique to the year the data was reported, making converting that information into local currency a laborious, though possible, task.

This lack in our ability to find GDP by state in local currency revealed the obstacle of navigating government databases. Most of our data was located through the Nigerian Bureau of Statistics (NBS), and to the best of our knowledge and ability during our case study, GDP by state in local currency was not found to be reported; however, at the time of the writing of this report, it was discovered that GDP by state in local currency was in fact published immediately prior to the performance of our case study, though only for 22/36 states. Whether the NBS website was updated to allow us to access this information at the time we conducted the study is unknown, but we do know that the report containing this information cannot be found if “State GDP” is used in the NBS search engine. It can only be found if “GDP” is used in the e-library search engine, and the viewer navigates through dozens of other reports. The specificity of search terms required to yield the desired information highlights the challenge of navigating government data sources, especially from government agencies with constrictive and unoptimized internal search engines.

Additionally, average income by state was another indicator that did not yield any information from national databases. This may warrant further investigation at the local government level, rather than relying on databases available at the national level. The complications this presents essentially compound the previously stated issues but with sufficient manual resources is a task that could be explored and accomplished.

Certain economic indicators were also found indirectly because the government briefs themselves did not directly or explicitly report the desired indicator. For example, “Population Employed Male” and “Population Employed Female” required us to calculate the indicator based on other different explicitly reported supplemental information, including percentages and region populations. This is particularly challenging because when moving forward and applying our methodology to other countries, this reveals that the explicitly reported information and the necessary calculations required to arrive at our chosen indicators will differ from country to country.

Another problem that is unique to Africa and particularly Nigeria is the unstable security conditions created by Boko Haram that have influenced the availability of data. As a result, the NBS has not released any data regarding living standards in the state of Borno. Though sample data was collected, it was not reported due to the fact that the sample was not representative since only households from accessible, safe areas could be reached. This indicates that sociopolitical conflicts present in countries or regions can influence the possibility of effective data aggregation.

We suspect that these variations and challenges will not be consistent across nations. Some countries will have more and different issues, while others will have less. This variability in itself will prove to be a complication in the future as access to certain data may not always be predictable. Moving forward in an effort to combat these challenges, we suggest FinMango maintain an adaptable and fluid approach to identifying diverse sources of data.

Conclusion

Using the set of economic indicators pre-established by FinMango, our team developed a data acquisition strategy consisting of:

1. Identifying relevant and up-to-date data sources, and
2. Creating and populating a database using Google Sheets.

We then applied our data aggregation strategy to examine the economic landscape in Nigeria. Through our case study of Nigeria, we found that the most recent and consistent data sources can be found through the national governments' statistical bureaus. Although economic data is widely available on these websites, there are still limitations to its use for our purpose. For instance, data may be inconsistent with our selected indicators, and data for certain states may be missing from the source. Due to time constraints, we were only able to conduct an in-depth economic case study of Nigeria, but there is potential to scale our strategy to extract economic data for other countries in Africa and across the globe. However, applying our strategy to other countries would be dependent on the data collection and presentation methods of other countries' statistical bureaus. After identifying relatively robust data sources, we hope that FinMango will use our work to collaborate with Google to automate the data extraction process to populate the databases more efficiently. Future research would also consist of identifying relevant and comprehensive sources to populate the healthcare datasets in order to gain a holistic overview of the socioeconomic landscape of countries as it relates to COVID-19.